

University Student Management System

Project Management Documentation — Planning, Scheduling, Communication & Risk Control

TEAM

Zomoroda (10 members)

TEAM LEADER

Yara Yasser Mostafa

PROJECT PERIOD

23 Sep – 30 Dec 2026

SUBMISSION DATE

11 October 2026

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1. Introduction & Tools Used

This document describes how Team Zomoroda plans, organizes and controls the development of the University Student Management System. It is prepared at the stage after the project idea was selected and before full coding starts, and it covers the four project management areas: **planning**, **scheduling**, **communication** and **risk control**.

Area	Tool	How the team uses it
Planning	Trello	Sprint lists, task cards with dates, members, labels, checklists and automation
Scheduling	Gantt chart (GanttProject file + PDF)	Timeline of the 7 sprints and all 148 tasks, milestones and progress
Communication	WhatsApp, Zoom, GitHub, face-to-face meetings	Daily coordination, online meetings, code discussion and key decisions
Version control	GitHub	Shared repository, branches, pull requests and commit history
Documentation	PDF	Proposal, sprint plan, team sheet and this project management document

2. Project Overview

Project Goal: Build a centralized web-based system for the main academic operations of a university, giving students, instructors, college administrators and university administrators role-based access to the functions they need.

SYSTEM ACTORS

- **Student** — registration, schedule, attendance, grades, GPA, transcript
- **Instructor** — courses, attendance, grades, academic advising
- **College Admin** — departments, courses, rules, exams, results, statistics
- **University Admin** — colleges, college admins, university statistics

TECHNOLOGY STACK

- **Frontend:** Angular & TypeScript
- **Backend:** .NET (REST APIs, authentication, validation)
- **Database:** Microsoft SQL Server
- **Version control:** Git & GitHub

MAIN MODULES (ONE OR TWO PER SPRINT)

Authentication & User Management

Student Registration

Instructor & Attendance

Grades, GPA & Transcript

Administration & Academic Management

Integration & Testing

3. Team Organization

No.	Team Member	Role
1	Yara Yasser Mostafa	Team Leader
2	Arwa Ayman Mohamed	Team Member
3	Esraa Saad Hamed	Team Member
4	Soha Abdelrady Adawy	Team Member
5	Moroge Mohamed Mahmoud	Team Member
6	Maryam Ahmed Hassan	Team Member
7	Maryam Ramadan Hassan	Team Member
8	Mennat Allah Mohamed Attallah	Team Member
9	Mona Mahmoud Hashem	Team Member
10	Yara Nehad Abu El Hassan	Team Member

DIVISION OF WORK

- **Backend & Database — 5 members:** server logic, database design and REST APIs
- **Frontend & UI/UX — 5 members:** user interface, client logic and responsive design
- Every Trello card is assigned to 2–4 members so no task depends on one person

RESPONSIBILITIES

- **Team Leader:** sprint planning, Trello board, final review and submission
- **Final reviewers:** Yara Yasser, Arwa Ayman, Mennat Allah
- **All members:** update their cards, attend meetings, follow the Definition of Done

4. Project Planning — Agile Sprint Plan

14 Weeks · 7 Sprints

The project follows an Agile approach with two-week sprints. Each sprint delivers one working module that is connected to the backend and the database. Sprint 5 (Administration) is four weeks long and is split into two parts on the Trello board: University Admin and College Admin.

Sprint	Duration	Dates	Main Focus	Tasks	Progress
Sprint 0	Week 1	23 Sep – 29 Sep	Planning, Requirements, Analysis & Setup	14	100%
Sprint 1	Week 2–3	30 Sep – 13 Oct	Authentication & User Management	12	46%
Sprint 2	Week 4–5	14 Oct – 27 Oct	Student Registration	14	0%
Sprint 3	Week 6–7	28 Oct – 10 Nov	Instructor & Attendance	17	0%
Sprint 4	Week 8–9	11 Nov – 24 Nov	Grades, GPA & Transcript	20	0%
Sprint 5.1	Week 10–11	25 Nov – 09 Dec	Administration — University Admin	15	0%
Sprint 5.2	Week 12–13	09 Dec – 22 Dec	Administration — College Admin	27	0%
Sprint 6	Week 14	23 Dec – 30 Dec	Integration, Testing & Finalization	29	0%

DEFINITION OF DONE

A sprint is considered done when all of the following conditions are satisfied:

- The implemented feature is completed.
- Frontend is connected to the Backend.
- Backend is connected to the Database.
- Validation is implemented.
- Errors are handled.
- The feature has been tested.
- The feature is integrated with the existing system.
- Code is committed and pushed to GitHub.
- The feature meets the defined requirements.

5. Task Management — Trello Board

[Public board](#)

Board: "University Student Management System" — public board; the link is in [Project-Management \(Trello\).md](#). All 10 team members are on the board.

5.1 BOARD STRUCTURE

List	Type	Cards	Purpose
Project Planning & Analysis	Sprint backlog	14	Task cards of one sprint module
Authentication & User Management	Sprint backlog	9	Task cards of one sprint module
Student Registration Module	Sprint backlog	14	Task cards of one sprint module
Instructor & Attendance Management	Sprint backlog	17	Task cards of one sprint module
Grade,GPA and Transcripts	Sprint backlog	20	Task cards of one sprint module
Administration & Academic Management (University/System Admin)	Sprint backlog	15	Task cards of one sprint module
Administration & Academic Management (College Admin)	Sprint backlog	27	Task cards of one sprint module
System Integration, Testing & Finalization	Sprint backlog	29	Task cards of one sprint module
To Do	Workflow	6	Sprint cards for upcoming sprints
In Progress	Workflow	1	Sprint card for the current sprint
Done	Workflow	1	Completed sprint cards (green "Sprint Completed" label)
Overdue cards	Workflow	3	Task cards that passed their due date (moved automatically)

5.2 CARD STRUCTURE

- **Title** — one task from the sprint plan
- **Start and due dates** — used to build the Gantt chart
- **Members** — the 2–4 people responsible for the task
- **Labels** — task type and priority
- **Technical Task Checklist** — the sub-steps of the task; progress is measured from the checked items

5.3 LABELS

Label	Color	Used for
Feature		Tasks that implement a new system feature
UI/UX		Interface design, layouts and visual components
Integration		Connecting modules, frontend, backend, database and APIs
Testing		Testing and verification tasks
Documentation		Documentation, manuals, guides and presentation
Bug Fixing		Identified errors that need fixing
High Priority		Needs immediate or important attention
Medium Priority		Normal priority
Low Priority		Can wait until higher-priority tasks are done
Sprint Completed		Added automatically when a sprint card is done

5.4 AUTOMATION RULES

Automation	Trigger	Action
Complete Card Checklists	A card is marked complete	All checklist items on the card are checked
Weekly Team Review	Every Saturday	A weekly report of the project cards is generated
Due Date Reminder	One day before a card is due	A reminder comment is posted on the card
Overdue Cards	A card becomes overdue	A comment tags the board and the card moves to the top of "Overdue Cards"
Sprint Completion	A sprint card in To Do / In Progress becomes due	The card moves to Done, gets the green label and a comment asks for the sprint review
Report Bug (card button)	A member clicks the button	The red "Bug Fixing" label is added and the board is notified

6. Project Scheduling — Gantt Chart

23 Sep – 30 Dec 2026

The Gantt chart was built from the start and due dates of the Trello cards. Each sprint is a summary bar; the darker part shows the completed share as of the status date. The full chart with every task is attached as a separate file (one page per sprint) together with the GanttProject source file.



6.1 MILESTONES

Date	Milestone
11 Oct 2026	Deliverable: Project Management (Trello + Gantt)
30 Dec 2026	Final presentation & system demo
End of each sprint	Sprint review; the Trello sprint card moves to Done automatically

6.2 SCHEDULING RULES

- Each sprint starts the day after the previous sprint ends (finish-to-start).
- Tasks inside a sprint are ordered by their dependencies: pages first, then logic, then testing at the end of the sprint.
- Every module has its own testing task in the last days of its sprint.
- Dates are changed only on Trello; the Gantt chart is updated from the board.

7. Communication Plan

Method	Purpose	Expected response / attendance
WhatsApp	Daily updates, urgent matters and team coordination	Respond within 3 hours
GitHub	Code collaboration, project tasks and issue tracking	Follow up within 24 hours
Zoom Meetings	Online team meetings and virtual sync-ups	Attend on time; notify in advance if absent
Face-to-Face Meetings	In-person discussions and key project decisions	Attend on time; notify in advance if absent
Trello	Task status, comments on cards, automatic reminders	Update your cards as soon as status changes

MEETINGS & REPORTING

- **Weekly Team Review** every Saturday (automatic Trello report)
- **Sprint review** at the end of each sprint
- **Daily updates** in the WhatsApp group
- Decisions are written on the related Trello card

ESCALATION PATH

1. The member raises the problem in the WhatsApp group
2. Members on the same card try to solve it within 24 hours
3. The Team Leader decides on reassignment or scope
4. The course instructor is contacted when needed

SUBMITTING ASSIGNMENTS

- **Internal deadline:** each assignment is completed and reviewed at least one day before the official deadline.
- **Final reviewers:** Yara Yasser Mostafa, Arwa Ayman Mohamed, Mennat Allah Mohamed Attallah.
- **Final submitter:** Yara Yasser Mostafa — Team Leader.

8. Risk Management

12 Risks

Approach: Identify the risks early, score each one by probability and impact, plan how to prevent it and what to do if it happens, then review the list every week.

RISK SCORING

- **Probability** and **Impact** are scored from 1 (Low) to 3 (High).
- **Risk score** = Probability × Impact.
- **6–9 High** act now and track weekly
- **3–4 Medium** mitigation planned and monitored
- **1–2 Low** accepted and watched

PROBABILITY ↑ / IMPACT →

High (3)	3 R04	6 R01 R02 R03
Medium (2)	2 R05 R06 R07 R09 R11	4 R08
Low (1)	1 R12	2 R10
	Low (1)	Medium (2) High (3)

8.1 RISK REGISTER

ID	Risk	Category	P	I	Score	Owner
R01	Schedule delay: tasks finish after their due dates and push later sprints	Schedule	3	3	9 · High	Team Leader
R02	Integration problems between the Angular frontend and the .NET API (different data formats, endpoints or auth handling)	Technical	3	3	9 · High	Backend Team
R03	Sprint 6 is overloaded: 29 integration, testing and documentation tasks in one week	Schedule	3	3	9 · High	Team Leader
R04	Overlap with university exams and other coursework (midterms and finals)	People	3	2	6 · High	Team Leader
R05	A team member becomes unavailable (illness, personal reasons, withdrawal)	People	2	2	4 · Medium	Team Leader
R06	Skill gaps in Angular, .NET or SQL Server slow down development	Technical	2	2	4 · Medium	All Members
R07	Requirement changes or scope creep after the proposal was approved	Scope	2	2	4 · Medium	Team Leader
R08	Late changes to the database design (ERD) affect many modules	Technical	2	3	6 · High	Backend Team
R09	Errors in complex logic: GPA calculation, credit-hour limits, schedule conflict detection	Quality	2	2	4 · Medium	Backend Team
R10	Security flaws in authentication or role-based access (a user sees pages of another role)	Quality	1	3	3 · Medium	Backend Team
R11	Merge conflicts or lost code on GitHub	Technical	2	2	4 · Medium	All Members
R12	Communication gaps: updates missed or decisions not shared with all members	Communication	1	2	2 · Low	Team Leader

8.2 RISK RESPONSES

ID	Mitigation (prevention)	Contingency (if it happens)	Trigger / warning sign
R01	Daily progress updates on WhatsApp; Trello due-date reminder and overdue automations; at least two members on each card; sprint scope reviewed every Saturday.	Move overdue cards to the top of the Overdue Cards list, reassign or pair up, and pull low-priority tasks into a later sprint.	A card reaches its due date without being marked complete (already 2 cards in Sprint 1).

ID	Mitigation (prevention)	Contingency (if it happens)	Trigger / warning sign
R02	Agree on API contracts (endpoints, request and response models) before each sprint; document them with Swagger; integrate and test at the end of every sprint, not only in Sprint 6.	Frontend and backend pairs fix the mismatch together in a joint session; use mock data on the frontend until the endpoint is ready.	API calls fail or return unexpected data during sprint testing.
R03	Test each module inside its own sprint; write test cases and documentation during Sprints 2–5; keep a running bug list on Trello.	Prioritise must-have workflows for the demo; move remaining polish items to after the demo.	More than 5 open bugs or untested modules at the start of Sprint 6.
R04	Mark exam weeks on the Gantt chart; plan lighter workloads in those weeks and front-load work before them.	Re-balance tasks across members and use the buffer days at the end of the sprint.	Exam timetable published or members report low availability.
R05	Every card has 2–4 members; code and decisions are documented on GitHub and Trello so others can continue.	Reassign the member's cards within 24 hours and inform the instructor if the absence is long.	No response within the agreed 3 hours on WhatsApp / 24 hours on GitHub.
R06	Short learning sessions in Sprints 0–1; pair less-experienced members with stronger ones; assign tasks by strength.	Ask for help in the team group or from the course instructor; simplify the implementation of the feature.	A task stays in progress much longer than its planned duration.
R07	Use the approved proposal and sprint plan as the scope baseline; every change is discussed with the Team Leader before it enters the backlog.	Estimate the change, add it as a new Trello card and move lower-priority work if needed.	New feature requests from the instructor or from team members.
R08	Review and freeze the ERD before Sprint 2; manage schema changes with migrations; one person approves schema changes.	Apply the change through a migration and update the affected APIs and pages in the same sprint.	A feature needs a table or column that is not in the ERD.
R09	Write the academic rules as test cases before coding; unit-test the calculation logic; review the results with sample data.	Fix the logic and re-run all related test cases before merging.	A test case or manual check gives a wrong GPA, total or conflict result.
R10	Check permissions on the server for every API, not only in the UI; test each role in Sprint 1 and again in Sprint 6.	Block the affected endpoint, fix the permission check and re-test all roles.	A role can open a page or API that should be blocked.
R11	Follow the agreed Git branching strategy; small pull requests reviewed before merging; protect the main branch.	Resolve conflicts together with the authors; restore from the repository history.	Failed merges or a broken main branch after a pull request.
R12	Agreed response times; weekly team review every Saturday; decisions recorded on the related Trello card.	Hold a short Zoom call to align everyone and update the cards.	Two members work on the same task, or a task is blocked without anyone knowing.

RISK MONITORING

- The risk register is reviewed in the Weekly Team Review every Saturday.
- Scores are updated when a trigger appears; new risks are added with the next free ID.
- High risks are discussed at every sprint review and assigned an owner.
- The Overdue Cards list on Trello is the main early warning for schedule risks.

9. Monitoring & Control

Status: 10 Oct 2026

CURRENT SPRINT

Sprint 1 · Week 2–3

TASKS COMPLETED

20 of 148

SPRINT 1 PROGRESS

46%

OVERDUE TASKS

2

- Progress tracking:** cards and checklist items completed on Trello; sprint cards move from To Do to In Progress to Done.
- Schedule control:** the Gantt chart is compared with the board every week; late cards are moved to Overdue Cards automatically.
- Current status:** Sprint 0 is complete (14 of 14 tasks). In Sprint 1, 2 tasks are overdue: "Create the Welcome Screen" (due 04 Oct), "Create the main application layout" (due 07 Oct). They are assigned and planned to finish before the end of the sprint on 13 Oct.
- Quality control:** the Definition of Done is checked before a card is completed; every sprint ends with a testing task.
- Change control:** new requirements become new Trello cards after the Team Leader approves them.

10. Attached Files & Links

Item	File / Link
Trello board (link)	Project-Management (Trello).md
Agile Sprint Plan	University Student Management System - Agile Sprint Plan.pdf
Gantt chart (all tasks)	USMS - Project Gantt Chart.pdf